

Mensik Pset 3

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Question 1

a

A standard error is a statistic that characterizes the amount of sampling variability (measured with the standard deviation) of the sampling distribution of estimates, *across* hypothetical iterations of an experiment. It is calculated by taking the summed squared difference of each iterations' ATE from the average ATE, dividing by the number of experiments (ways of randomizing treatment), and the square root.

Within a given experiment, variability in outcomes is measured with standard deviation. Like SE, it is the sum of the squared difference between each observation (in this case) and the average, divided by the number of observations or one less than that number. It denotes dispersion of observations from the mean for the sample.

SD usually is about dispersion in an observed variable.

b

It would be nice to know the treated and untreated outcome for each subject in an experiment. This would be the trouble with this is that The sharp null hypothesis is that the treatment effect for all subjects in an experiment is zero. This is helpful because Randomization Inference is used to test the sharp null by finding the p-values for many or all possible random assignments.

Randomization Inference A secondary problem of causal inference is we cannot derive test statistics from unobserved potential outcomes. The sharp null is helpful in this respect: stating that the treatment effect is zero for all units also means that unobserved and observed potential outcomes are the same.

Using that 'complete' battery, we can simulate randomization and produce a 'true'-under-the-null sampling distribution for estimate ATEs, and derive the p-value, or probability of obtaining an ATE as large as that obtained in the actual experiment in question, were the treatment effect nil.

c

Imposing a 95% confidence interval around an estimator creates an estimated interval that will contain the "truth" 95% of the time, under repeated sampling.

d

Difference between complete random assignment and cluster random assignment. In complete random assignment, the number of those in the treatment (m) and control ($N-m$) is known and the probability for each unit of assignment to treatment is m/N .

In expectation, this creates balance for covariates.

Block random assignment, in effect, turns a single experiment into some number of mini-experiments: study units are divided into blocks or strata before treatment is assigned within each block.

In cluster random assignment, subjects likely to have similar outcomes are put into groups. Treatment then is randomized across groups – if a cluster is a family for example, the family becomes the unit of observation rather than individual family members.

e

Balanced designs should allow confidence intervals to be “conservative” by reducing sampling variability at least when the variances of outcomes are similar within treatment conditions.

Question 4

Sub	Y0	YA	YB
miriam	1	2	3
ben	2	3	3
helen	3	4	3
eva	4	5	3
billie	5	6	3

treatment effects defined as $YA - Y0$ or $YB - Y0$

a.

Which treatment, A or B, will generate a sampling distribution with a smaller standard error

```
schedule <- data.frame("Y(0)" = 1:5, "Y(A)" = 2:6, "Y(B)" = rep(3,5))
```

Treatment A's standard error for the ATE is:

```
ta_se <- sqrt((1/4)*((3*(var(schedule$Y.0))) / (2) +  
                    (2*(var(schedule$Y.A))) / (3) +  
                    (2*cov(schedule$Y.0, schedule$Y.A))))  
print(ta_se)
```

```
## [1] 1.613743
```

Treatment B's standard error for the ATE is:

```
tb_se <- sqrt((1/4)*((3*(var(schedule$Y.0))) / (2) +  
                    (2*(var(schedule$Y.B))) / (3) +  
                    (2*cov(schedule$Y.0, schedule$Y.B))))  
print(tb_se)
```

```
## [1] 0.9682458
```

Treatment B then has a smaller standard error. This might also have been inferable because it has a variance of 0 (all of $Y(B) = 3$).

b.

Treatment B can be considered a “Robin Hood” treatment: one that “steals” outcomes from the “rich” (those with a high pre-treatment or untreated outcome) by lowering their treated outcomes while “giving” (raising treated outcomes) for those with low pre-treatment or untreated outcomes, the “poor.”

This minimizes variance in treated potential outcomes and the covariance between treated and untreated potential outcomes, and lowers sampling variability, so shrink standard errors in ways that may overstate certainty.

Question 5

Using table 2.1, imagine experiment allocates one village to treatment

```
getwd()
```

```
## [1] "/Users/kristinamensik/Documents/GitHub/regression-study/Experiments"
```

```
t2_1 <- read_csv("/Users/kristinamensik/Documents/GitHub/regression-study/Experiments/doi-10.60600-yu-v
```

```
## Rows: 7 Columns: 4
## -- Column specification -----
## Delimiter: ","
## dbl (4): Village i, Yi[0], Yi[1], Treatment Effect
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
View(t2_1)
```

a

Calc estimated DIM for all 7 possible randomizations

```
set.seed(1090903)

# assignment procedure

ap_q5 <- declare_ra(N = nrow(t2_1), m = 1)

# randomizations permutation matrix

pm_q5 <- obtain_permutation_matrix(ap_q5)

# results vector

des_q5 <- c()

# difference-in-means, all permutations

for (i in 1:ncol(pm_q5)) {
```

```

t2_1$ta <- pm_q5[, i] #treatment assignment

treated_mean <- mean(
  t2_1$`Yi[1]`[t2_1$ta == 1]
)

control_mean <- mean(
  t2_1$`Yi[0]`[t2_1$ta == 0]
)

dim <- treated_mean - control_mean
des_q5 <- c(des_q5, dim)
}

print(des_q5)

```

```
## [1] 15.0000000  0.0000000  4.1666667  0.8333333 15.8333333  0.0000000 -0.8333333
```

b

The average across estimates is:

```
print(mean(des_q5))
```

```
## [1] 5
```

The true treatment effect is the average of treatment effects for each unit.

```

true_ate <- mean(t2_1$`Treatment Effect`)

print(true_ate)

```

```
## [1] 5
```

Both the average across estimates and the true ATE are 5.

c

Standard deviation of the 7 estimates is identical to the standard error implied by equation 3.4

```

se_q5 <- sqrt((1/6)*
  ((1*(var(t2_1$`Yi[0]`))) / (6) +
  (6*(var(t2_1$`Yi[1]`))) / (1) +
  (2*cov(t2_1$`Yi[0]`,t2_1$`Yi[1]`))))

print(se_q5)

```

```
## [1] 7.296625
```

```
sd_eq5 <- sd(des_q5)
print(sd_eq5)
```

```
## [1] 7.296625
```

Both are 7.296.

d

Why this experimental design has more sampling variability than the design in which 2 villages out of 7 are assigned to treatment.

Equation 3.4 shows to weight the variance of treated potential outcomes by the ratio of untreated units to treated units. More treated units means a larger denominator and thus smaller standard error or sampling variability.

e

Having only one unit (village) assigned to treatment means that it is the only outcome informing the estimated treatment effect, so that effect will reflect particularities of that village. This is a continuation of the point in the previous answer on weighted variances.

- how distinct?

Question 6

Assume visas assigned with complete random assignment, conduct 10k simulated random assignments under the sharp null.

```
data <- read_dta("/Users/kristinamensik/Documents/GitHub/regression-study/Experiments/doi-10.60600-yu-v
view(data)

set.seed(1090903) #set seed

haj_declaration <- declare_ra(nrow(data), m = sum(data$success)) #ra procedure

haj_permmatrix <- obtain_permutation_matrix(haj_declaration) #sims matrix

haj_dim_estimates <- c() #init storage vector

for (i in 1:ncol(haj_permmatrix)) {
  data$ri <- haj_permmatrix[, i] #go thru every perm

  haj_treated_mean <- mean(data$views[data$ri == 1])
  haj_control_mean <- mean(data$views[data$ri == 0])

  haj_dim <- haj_treated_mean - haj_control_mean #calc dim
  haj_dim_estimates <- c(haj_dim_estimates, haj_dim)
```

```
}

#mean for sharp null should be 0:

print(round(mean(haj_dim_estimates)))
```

```
## [1] 0
```

How many generate an estimated ATE at least as large as actual estimates?

Actual ATE:

```
clingingsmith_dim <-
  mean(data$views[data$success == 1]) -
  mean(data$views[data$success == 0])

print(clingingsmith_dim)
```

```
## [1] 0.4748337
```

How many?

```
print(sum(haj_dim_estimates >= clingingsmith_dim))
```

```
## [1] 19
```

Implied one tailed p value is:

```
print(mean(haj_dim_estimates >= clingingsmith_dim))
```

```
## [1] 0.0019
```

How many simulated RAs generate an estimate ATE at least as large in absolute value?

```
print(sum(abs(haj_dim_estimates) >= abs(clingingsmith_dim)))
```

```
## [1] 34
```

Two-tailed p value:

```
print(mean(abs(haj_dim_estimates) >= abs(clingingsmith_dim)))
```

```
## [1] 0.0034
```

Question 7

Transformation, then test sharp null.

Potential outcomes schedule

```

Y0_6 <-c(1,0,0,4,3)
Y1_6<-c(2,11,14,0,3)
tformed_Y1 <- Y1_6 - 7
d_6 <- c(rep(0,length(Y0_6)), rep(1,length(tformed_Y1)))
null <- c(Y0_6, tformed_Y1)
sched_6 <- data.frame(cbind(null, d_6))

```

```

set.seed(1090903) #set seed

dec <- declare_ra(nrow(sched_6), m = sum(sched_6$d_6)) #ra procedure

permmat <- obtain_permutation_matrix(dec) #sims matrix

dims <- c() #init storage vector

for (i in 1:ncol(permmat)) {
  sched_6$ri <- permmat[, i] #go thru every perm

  treated_mean <- mean(sched_6>null[sched_6$ri == 1])
  control_mean <- mean(sched_6>null[sched_6$ri == 0])

  dim6 <- treated_mean - control_mean #calc dim
  dims <- c(dims, dim6)
}

```

Transformed outcome vector to get the experiment's ATE

```

exp_dim <- mean(sched_6>null[sched_6$d_6==1]) - mean(sched_6>null[sched_6$d_6==0])

print(exp_dim)

```

```
## [1] -2.6
```

```
print(mean(abs(dims) >= abs(exp_dim)))
```

```
## [1] 0.4126984
```

I used a two tailed test because a diet program may well cause weight gain instead of loss. The p-value produced is 0.412, so greater than 0.05, so we can't reject the sharp null that the program had no effect across participants.

Question 8

```

q8_text <- "https://hdl.handle.net/10079/10b7b9e8-ab68-493b-81ff-39063be637ae"

data <- read.csv(q8_text, header = TRUE, sep = ",")

```

a

Estimated average treatment effect of having a 2yr term on bill introductions:

```
tx <- data[data$txas0_arkansas1 == 0,] #tx block
ar <- data[data$txas0_arkansas1 == 1,] #ak block

tx_ate <- mean(tx$bills_introduced[tx$term2year == 1]) - mean(tx$bills_introduced[tx$term2year == 0])

ar_ate <- mean(ar$bills_introduced[ar$term2year == 1]) - mean(ar$bills_introduced[ar$term2year == 0])
```

For Texas, ATE is:

```
print(tx_ate)
```

```
## [1] -16.74167
```

For Arkansas, ATE is:

```
print(ar_ate)
```

```
## [1] -10.09477
```

b

Standard error of ests of ATE:

```
m_tx <- sum(tx$term2year)
m_ar <- sum(ar$term2year)

var0_tx <- var(tx$bills_introduced[tx$term2year == 0])
var1_tx <- var(tx$bills_introduced[tx$term2year == 1])

var0_ar <- var(ar$bills_introduced[ar$term2year == 0])
var1_ar <- var(ar$bills_introduced[ar$term2year == 1])

se_tx <- sqrt((var0_tx / (nrow(tx) - m_tx)) + (var1_tx / m_tx))
se_ar <- sqrt((var0_ar / (nrow(ar) - m_ar)) + (var1_ar / m_ar))
```

For Texas, the SE is:

```
print(se_tx)
```

```
## [1] 9.345871
```

For Arkansas, the SE is:

```
print(se_ar)
```

```
## [1] 3.395979
```

c

ATE combined?

```
#block weights
```

```
weight_tx <- nrow(tx) / nrow(data)
```

```
weight_ar <- nrow(ar) / nrow(data)
```

```
states_ate <- (weight_tx * tx_ate)+(weight_ar*ar_ate)
```

ATE combined is:

```
print(states_ate)
```

```
## [1] -13.2168
```

d

Why simply pooling data for the two states and comparing averages would lead to a biased ATE est.

Texas and Arkansas are samples of states – it's not clear whether they were randomly selected, so it's more likely that they were chosen because of some features that might predict the outcome of interest. We can only assume that randomization happens within the blocks. The probability of receiving treatment then is in respect to within each state, not across the states.

e

SE for the overall ATE is:

```
states_se <- sqrt(  
  ((se_tx^2)*(weight_tx^2)) + ((se_ar^2)*(weight_ar^2))  
)  
print(states_se)
```

```
## [1] 4.74478
```

f

RI to test the sharp null – equation 3.12: randomize within block, then standard errors within each block can then be combined using squared block.

```
#set seet  
nrow(tx)
```

```
## [1] 31
```

```
nrow(ar)
```

```
## [1] 35
```

```
m_tx
```

```
## [1] 15
```

```
m_ar
```

```
## [1] 18
```

15 treated senators in texas block out of 31, 18 treated senators in Ar block out of 35.

```
sim_tx <- sample(c(rep(1, m_tx), rep(0, nrow(tx) - m_tx)))
```

```
sim_ar <- sample(c(rep(1, m_ar), rep(0, nrow(ar) - m_ar)))
```

```
sim_tx_ate <- mean(tx$bills_introduced[sim_tx == 1]) -  
  mean(tx$bills_introduced[sim_tx == 0])
```

```
sim_ar_ate <- mean(ar$bills_introduced[sim_ar == 1]) -  
  mean(ar$bills_introduced[sim_ar == 0])
```

```
states_ates <- (weight_tx*sim_tx_ate)+(weight_ar*sim_ar_ate)
```

```
set.seed(1090903)
```

```
B <- 1000
```

```
ates <- replicate(B, {  
  sim_tx <- sample(c(rep(1, m_tx), rep(0, nrow(tx) - m_tx)))  
  sim_ar <- sample(c(rep(1, m_ar), rep(0, nrow(ar) - m_ar)))  
  sim_tx_ate <- mean(tx$bills_introduced[sim_tx == 1]) -  
    mean(tx$bills_introduced[sim_tx == 0])  
  sim_ar_ate <- mean(ar$bills_introduced[sim_ar == 1]) -  
    mean(ar$bills_introduced[sim_ar == 0])  
  (weight_tx*sim_tx_ate)+(weight_ar*sim_ar_ate)  
})
```

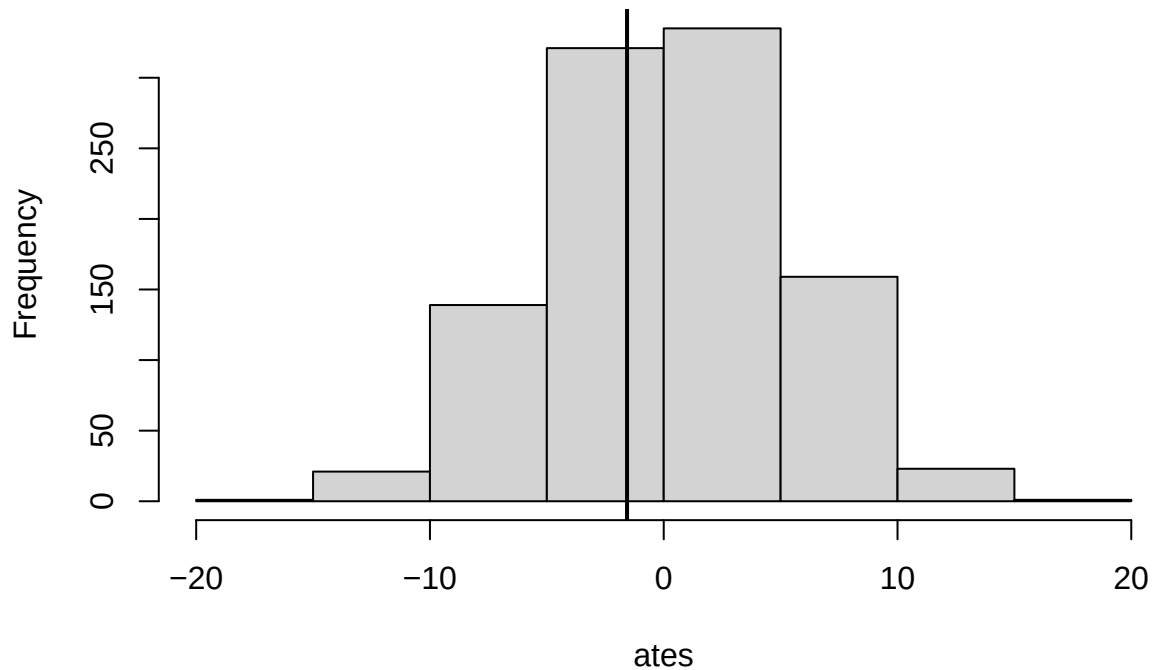
```
summary(ates)
```

```
##      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.  
## -15.3407  -3.2832    0.1898    0.2346    3.7694   17.5401
```

```
hist(ates)
```

```
abline(v = states_ates, lwd = 2)
```

Histogram of ates



```
p_states <- mean(
  abs(ates) >= abs(states_ate)
)
```

```
p_states
```

```
## [1] 0.007
```

A two-tailed test is appropriate here because there's no clear reason to expect the effect to be strictly negative or positive. The p value of 0.007 means we can reject the sharp null. ## Question 9 ### a The problem for non-interference is if betting on the treatment horse affects the outcomes for the non treatment horse. A workaround might be redefining $Y_i(0)$, so that it is bets for horse i when the other horse (horse ii ?) in the same race is treated ($Y_i(0)$ would be the standard bets on horse i when i treated). $Y_i(0)$ then records the potential outcome when d_i is 0 and d_{ii} is 1 – potential outcomes are the same in this case no matter the assignment of other horses.

b

do treatment and control horses attract similarly sized bets prior to intervention; use randomization inference to test the sharp null hyp that bets have no affect.

```
data <- read.csv("/Users/kristinamensik/Documents/GitHub/regression-study/Experiments/Camerer_JPEsubset.
View(data)

set.seed(1090903)

horse_blocks <- declare_ra(
```

```

N = nrow(data),
blocks = data$pair,
m = 1)

permmatrix <- obtain_permutation_matrix(horse_blocks)

ests_vector <- c()

for (i in 1:ncol(permmatrix)) {
  data$ri <- permmatrix[, i]

  pre_treat <- mean(data$preexperimentbets[data$ri == 1])

  pre_treat_control <- mean(data$preexperimentbets[data$ri == 0])

  ests_vector[i] <- pre_treat - pre_treat_control
}

```

Is mean of pre-experiments ATEs 0?

```
print(round(mean(ests_vector)))
```

```
## [1] 0
```

Actual pre experiment bets DIM:

```
pre_dim <- mean(data$preexperimentbets[data$treatment==1]) - mean(data$preexperimentbets[data$treatment
print(pre_dim)
```

```
## [1] -15.29412
```

Sharp null p value:

```
print(mean(abs(ests_vector) >= abs(pre_dim)))
```

```
## [1] 0.3622
```

The p-value of 0.6312 prohibits us from rejecting the sharp null (the sharp null being: no difference in pre-experiments for the treated horses vs control horses).

This means there are differences in pre-experiment bets for the horses assigned to control and to treatment. As a randomization check, this failure to reject the null confirms randomization.

(c)

Average increase in bets during the experimental period for treatment and control horses.

```
dim_ate <- mean(data$experimentbets[data$treatment == 1]) - mean(data$experimentbets[data$treatment == 0])
print(dim_ate)
```

```
## [1] -110.1765
```

On average, the total amount betted on horses assigned to treatment was \$110 less than the total for control horses, in the experimental period. This backs up the researcher's supposition that betting on long shot horses would diminish odds to decline.

(d)

The ATE was:

```
print(dim_ate)
```

```
## [1] -110.1765
```

and done this way it is:

```
pairwise <- data %>%
  difference_in_means(experimentbets ~ treatment, ., pair) %>% coefficients
print(pairwise)
```

```
## treatment
## -110.1765
```

They are the same.

e

Shap null for bets on treated horse for any horse

```
set.seed(1090903)

dims <- c()

for(i in 1:ncol(permmatrix)) {
  data <-
    data %>%
    mutate(var = permmatrix[, i])

  dims[i] <-
    data %>%
    difference_in_means(experimentbets ~ var, ., pair) %>%
    coefficients
}
```

test sharp null – mean is 0?

```
print(round(mean(dims)))
```

```
## [1] -2
```

hmm.

```
print(mean(abs(dims) >= abs(dim_ate)))
```

```
## [1] 0.3152
```

I get a p-val of 0.31, far from the 0.05 standard. This suggests I can't reject the sharp null and that 31% of the time we won't observe the truth – the 110 dollars may be such an observation.

Question 10

Classrooms of 25 apiece, 800 students, classrooms themselves assigned randomly as clusters to treatment and control (block clustering).

In 3.22, we divide by $k - 1$ – one less than the total number of clusters – rather than by $N - 1$. Instead of unit-level potential outcomes, we use the average of potential outcomes at the cluster-level.

In this example, the denominator will be smaller than if the experimenter just did complete random assignment, which would produce larger standard errors for the ATE. However, the numerator will also be smaller. That's because of cluster means: 25 students per cluster means the cluster-level variance of outcomes will be 1/25th of individual-level variance. In sum, this could produce similar standard errors as would be produced by complete random assignment.

Question 11

```
table3<-read_csv("/Users/kristinamensik/Documents/GitHub/regression-study/Experiments/GerberGreenBook_Cluster")
```

a

```
## Rows: 14 Columns: 4
## -- Column specification -----
## Delimiter: ","
## dbl (4): Village, Y, D, Block
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
view(table3)
```

```
names(table3)[2] <- "Y0"
names(table3)[3] <- "Y1"
```

```
table3 <- table3 %>%
```

```

mutate(Cluster1 = rep(1:(n() / 2), each = 2)) #clusters

set.seed(1090903)

table3_declaration <-
  declare_ra(nrow(table3), clusters = table3$Cluster1, m = 4) # 4 clusters, cluster randomization

table3 <-
  table3 %>%
  mutate(D1 = conduct_ra(table3_declaration))

cluster_means_t3 <- table3 %>% # cluster means table
  group_by(Cluster1) %>%
  summarise(
    Y0bar1 = mean(Y0),
    Y1bar1 = mean(Y1)
  )

var_Y01 <- var(cluster_means_t3$Y0bar1) #var y0
var_Y11 <- var(cluster_means_t3$Y1bar1) # var y 1
cov_Y01_Y11 <- cov(cluster_means_t3$Y0bar1, cluster_means_t3$Y1bar1)

num_clusters <- length(unique(table3$Cluster1))
num_treated <- sum(table3$D1) / 2

table3_se <- sqrt(
  (1 / (num_clusters - 1)) * (
    (num_treated / (num_clusters - num_treated)) * var_Y01 +
    ((num_clusters - num_treated) / num_treated) * var_Y11 +
    2 * cov_Y01_Y11
  )
)

print(table3_se)

## [1] 5.083268

```

```

new_clusters <- rep(NA, 14) # making new clusters

pairs <- list(
  c(1,14),
  c(2,13),
  c(3,12),
  c(4,11),
  c(5,10),
  c(6,9),
  c(7,8)
)

```

```

for (i in seq_along(pairs)) {
  new_clusters[pairs[[i]]] <- i
}

table3$Cluster2 <- new_clusters

set.seed(1090903)

q11b_declaration <-
  declare_ra(nrow(table3), clusters = table3$Cluster2, prob = 0.5) #cluster randomization, send half of

table3 <-
  table3 %>%
  mutate(D2 = conduct_ra(q11b_declaration))

cluster_means2 <- table3 %>% #means
  group_by(Cluster2) %>%
  summarise(
    Y0_bar2 = mean(Y0),
    Y1_bar2 = mean(Y1)
  )

var_Y02 <- var(cluster_means2$Y0_bar2) #var
var_Y12 <- var(cluster_means2$Y1_bar2)

cov_Y02_Y12 <- cov(cluster_means2$Y0_bar2, cluster_means2$Y1_bar2)

numclusters2 <- length(unique(table3$Cluster2)) #count
numtreated2 <- sum(table3$D2)/2

table3_se2 <- sqrt(
  (1 / (numclusters2 - 1)) * (
    (numtreated2 / (numclusters2 - numtreated2)) * var_Y02 +
    ((numclusters2 - numtreated2) / numtreated2) * var_Y12 +
    2 * cov_Y02_Y12
  )
)

print(table3_se2)

```

b

```
## [1] 1.054877
```

```
cov_Y01_Y11
```

```
## [1] 35.61905
```

```
cov_Y02_Y12
```

```
## [1] 0.702381
```

(c) Why different SEs? Implications of design for cluster randomized experiments?

Homogeneity within clusters shapes the variance of cluster-level means, and so leads to different standard errors. This is evidenced by the variance of the first cluster (35.6) being about 5 times the variance of method number two (0.70) – these were products of clusters more homogenous in the first than second go-around.

Assigning clusters is often necessitated by contexts of the field in question that make interference likely, and are out of the researchers control. Further still, groups that are likely interference/cluster candidates are probably often homogenous in some way that makes it hard for the researcher to avoid increased standard errors if they are to protect inference. Net net then, to the degree researchers can avoid homogeneity within clusters, that seems good, but otherwise it is important just to be mindful of in interpreting standard errors.

Question 12

a

```
table12 <- data.frame(  
  School = c("A", "B", "B", "C", "C", "C"),  
  Room = c("A1", "B1", "B2", "C1", "C2", "C3"),  
  Y0 = c(0,0,0,0,0,0),  
  Y1 = c(0,1,1,2,2,2)  
)  
  
ATE <- mean(table12$Y1) - mean(table12$Y0)
```

The ATE among the 6 classrooms is:

```
print(ATE)
```

```
## [1] 1.333333
```

b

3 possible randomizations: one where A is in treatment, one where B is in treatment, one where C is in treatment.

```
a_treat <- mean(table12$Y1[table12$School == "A"])  
b_treat <- mean(table12$Y1[table12$School == "B"])  
c_treat <- mean(table12$Y1[table12$School == "C"])  
  
arrangements_dim <- (a_treat + b_treat + c_treat) / 3
```

ATE when A is in treatment is:

```
print(a_treat)
```

```
## [1] 0
```

ATE when B is intreatment is:

```
print(b_treat)
```

```
## [1] 1
```

ATE when C is intreatment is:

```
print(c_treat)
```

```
## [1] 2
```

The question asks if the DIM is unbiased. We know that the true ATE is 1.33. In expectation the DIM is

```
print(arrangements_dim)
```

```
## [1] 1
```

So the DIM is biased down, which might reflect a covariance between cluster size and potential outcomes.

c

When school A and school B are combined for purposes of random assignment so that there's a .5 prob that either school c is placed in treatment or schools A and B are placed in treatment

```
ab_treat <- mean(table12$Y1[table12$School=="A"|table12$School == "B"])
```

```
c_treat_c <- mean(table12$Y1[table12$School == "C"])
```

```
grand_dim <- (ab_treat + c_treat_c) / 2
```

. . . we get a DIM of

```
print(grand_dim)
```

```
## [1] 1.333333
```

. . . that suggests the design set up unbiased estimates. It did so by overcoming disparate cluster sizes by balancing the number of units observed in treatment and in control. Ideally, researchers would work with more plentiful clusters, and should try to ensure that clusters are the same size from the jump; if they cannot, posthoc probability weighting should be used.